

Few-shot 3D Point Cloud Semantic Segmentation

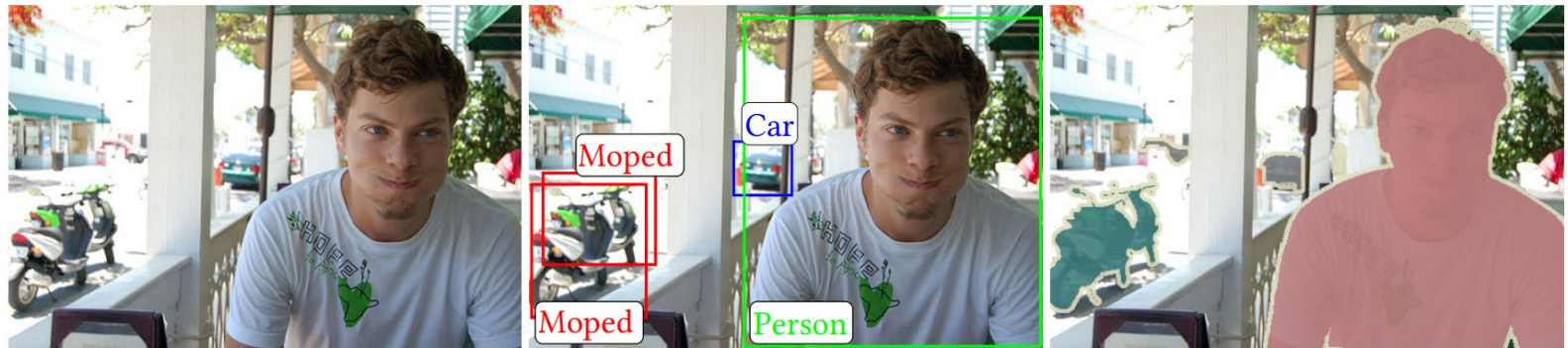
Yun Liu

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- Zhaochong An, Guolei Sun*, **Yun Liu***, Fayao Liu, Zongwei Wu, Dan Wang, Luc Van Gool, and Serge Belongie. “Rethinking Few-shot 3D Point Cloud Semantic Segmentation”. **IEEE CVPR, 2024.**
- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Min Wu, Ming-Ming Cheng, Ender Konukoglu, and Serge Belongie. “Multimodality Helps Few-Shot 3D Point Cloud Semantic Segmentation”. **ICLR 2025.**
- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

Image Semantic Segmentation

- Training fully-supervised image semantic segmentation models requires large-scale datasets with pixel-wise annotations. However, creating and labeling such datasets demands substantial resources.
- Few-shot semantic segmentation learns to segment target classes (novel classes) in the *query* image using only a few pixel-wise annotated *support* images, enabling segmentation models trained on *base* classes to generalize to *novel* classes.



(a) Original image

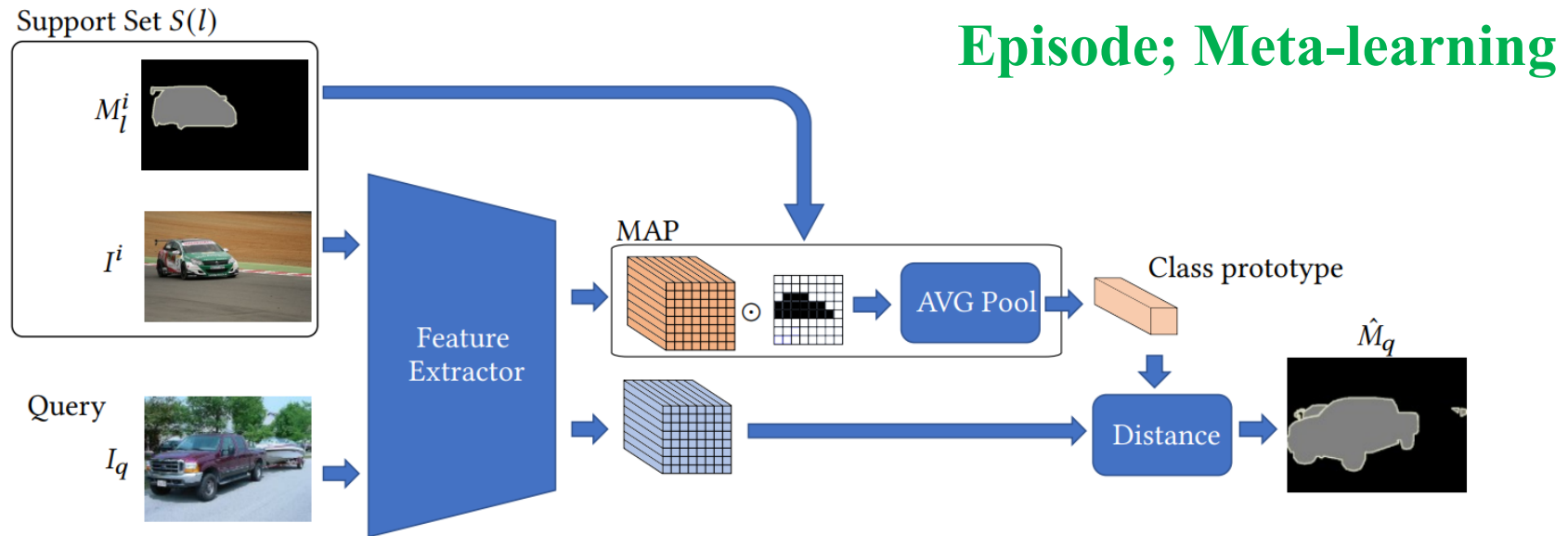
(b) Object detection

(c) Semantic Segmentation

- Nico Catalano, and Matteo Matteucci. “Few Shot Semantic Segmentation: a review of methodologies, benchmarks, and open challenges”. arXiv preprint arXiv:2304.05832 (2023).

Few-shot Image Semantic Segmentation

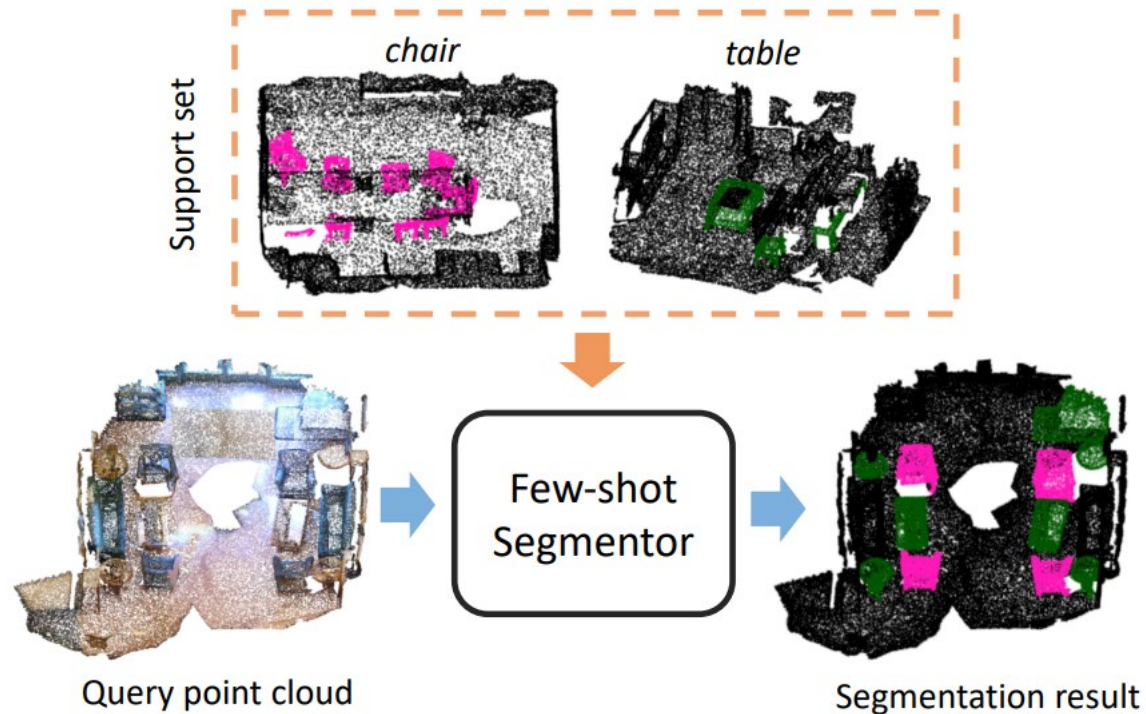
- A shared feature extractor gets a feature volume from both the support set and query images. The **Masked Average Pooling (MAP)** module takes the feature volume from the support set and masks its ground truth with the Hadamard product $\odot$ to compute the class prototype.
- The prediction mask $\hat{M}_q$ is calculated as a **metric** between the vector at each spatial location in the query feature volume with the class prototype.



- Nico Catalano, and Matteo Matteucci. “Few Shot Semantic Segmentation: a review of methodologies, benchmarks, and open challenges”. arXiv preprint arXiv:2304.05832 (2023).

Few-shot 3D Point Cloud Semantic Segmentation

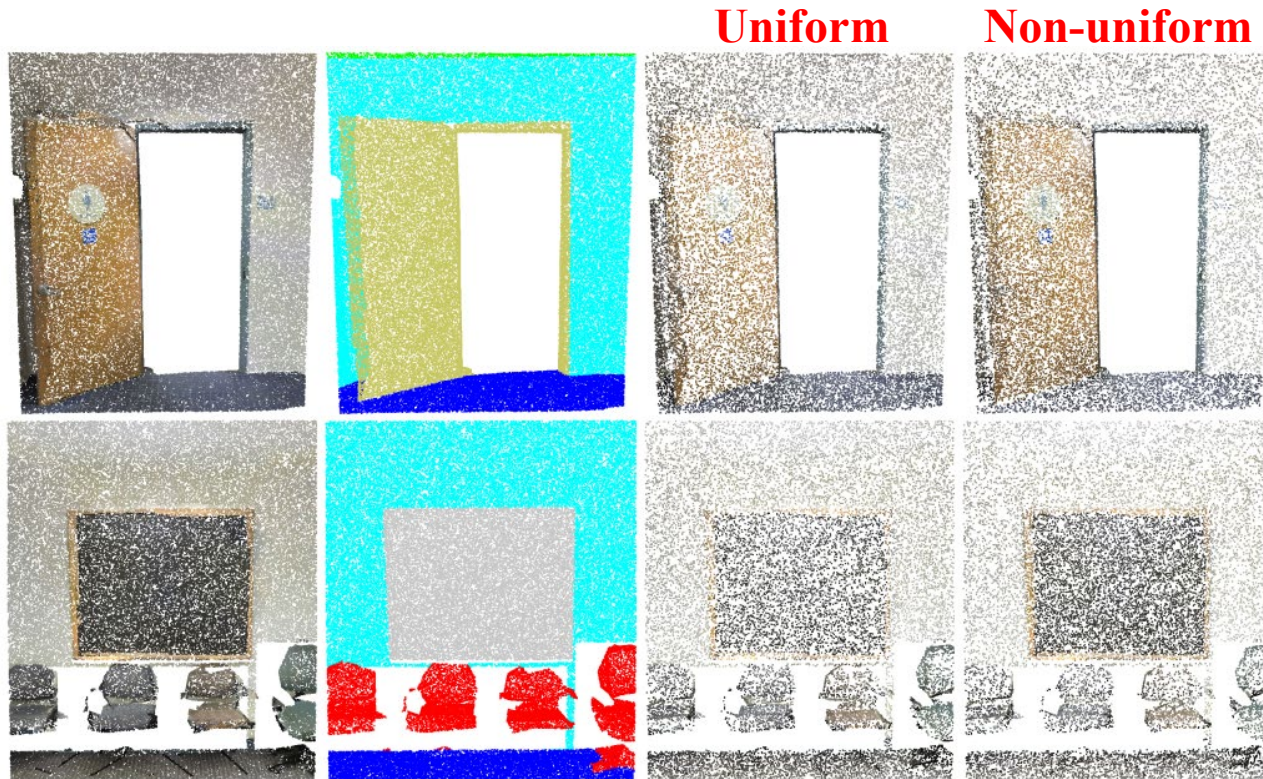
- Few-shot point cloud semantic segmentation (**FS-PCS**) learns to segment **target classes** in the *query* point cloud given a few annotated *support* point clouds. This figure illustrates an example with the 2-way 1-shot setting, which means that we have two target classes (chair and table) and one support point cloud for each class.



- Na Zhao, Tat-Seng Chua, and Gim Hee Lee. "Few-shot 3d point cloud semantic segmentation". In Proceedings of the IEEE/CVF CVPR, pp. 8873-8882. 2021.

Issue 1: Foreground Leakage

- The point sampling process in previous FS-PCS is non-uniform, favoring more points in the foreground than in the background. This leads to foreground leakage, a noticeable density bias toward foreground classes.



From left to right: (1) The original point cloud; (2) Ground truth of all categories; (3) Our corrected input with 20,480 points in a uniform distribution; (4) Input with 20,480 points in a biased distribution.

Issue 1: Foreground Leakage

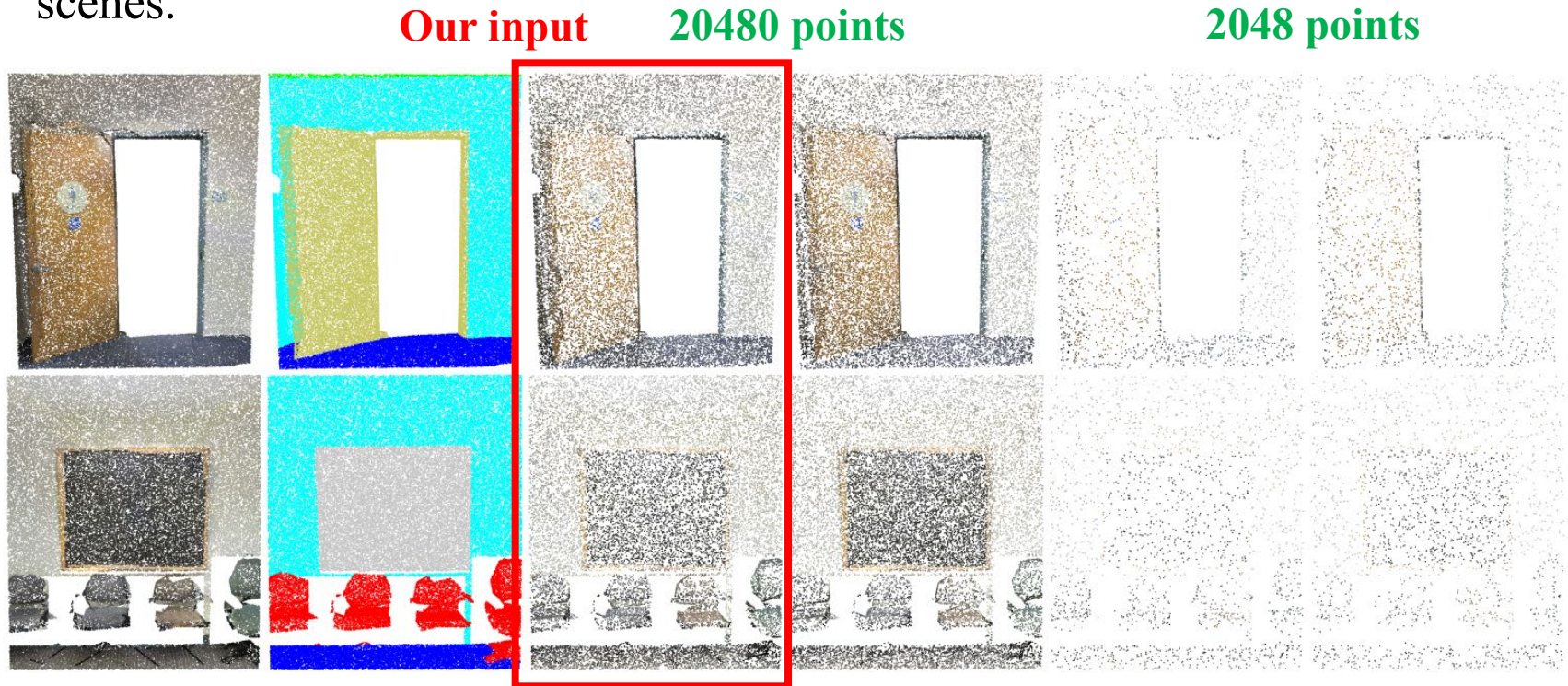
- The current non-uniform point sampling leads to a noticeable **point density disparity** between foreground and background, which induces models to segment foreground classes by identifying **denser** regions, instead of learning semantic knowledge transfer from support to query.
- Addressing this issue results in a significant performance drop in existing methods.

	Methods	1-shot (S3DIS)			5-shot (S3DIS)			1-shot (ScanNet)			5-shot (ScanNet)		
		S^0	S^1	mean	S^0	S^1	mean	S^0	S^1	mean	S^0	S^1	mean
<i>w/ FG</i>	AttMPTI [56]	64.89	66.15	65.52	76.56	83.08	79.82	62.14	58.65	60.39	68.79	68.66	68.73
	QGE [29]	74.05	73.61	73.83	74.65	83.21	78.93	63.50	57.61	60.56	70.72	65.68	68.20
	QGPA [11]	62.72	61.95	62.33	76.30	87.29	81.80	56.47	51.72	54.10	81.57	72.75	77.16
<i>w/o FG</i>	AttMPTI [56]	41.56	41.27	41.41	50.55	46.13	48.34	33.36	31.81	32.58	37.95	36.30	37.12
	QGE [29]	46.27	47.76	47.02	47.74	59.77	53.76	37.72	34.64	36.18	48.73	39.95	44.34
	QGPA [11]	35.62	41.13	38.38	43.54	47.50	45.52	40.03	35.54	37.78	46.17	42.24	44.20

Comparisons in the mIoU metric between *with* foreground leakage (*w/ FG*) and *without* foreground leakage (*w/o FG*) for existing methods. The results are for 1-way segmentation setting. S^0/S^1 refers to the i -th split for inference.

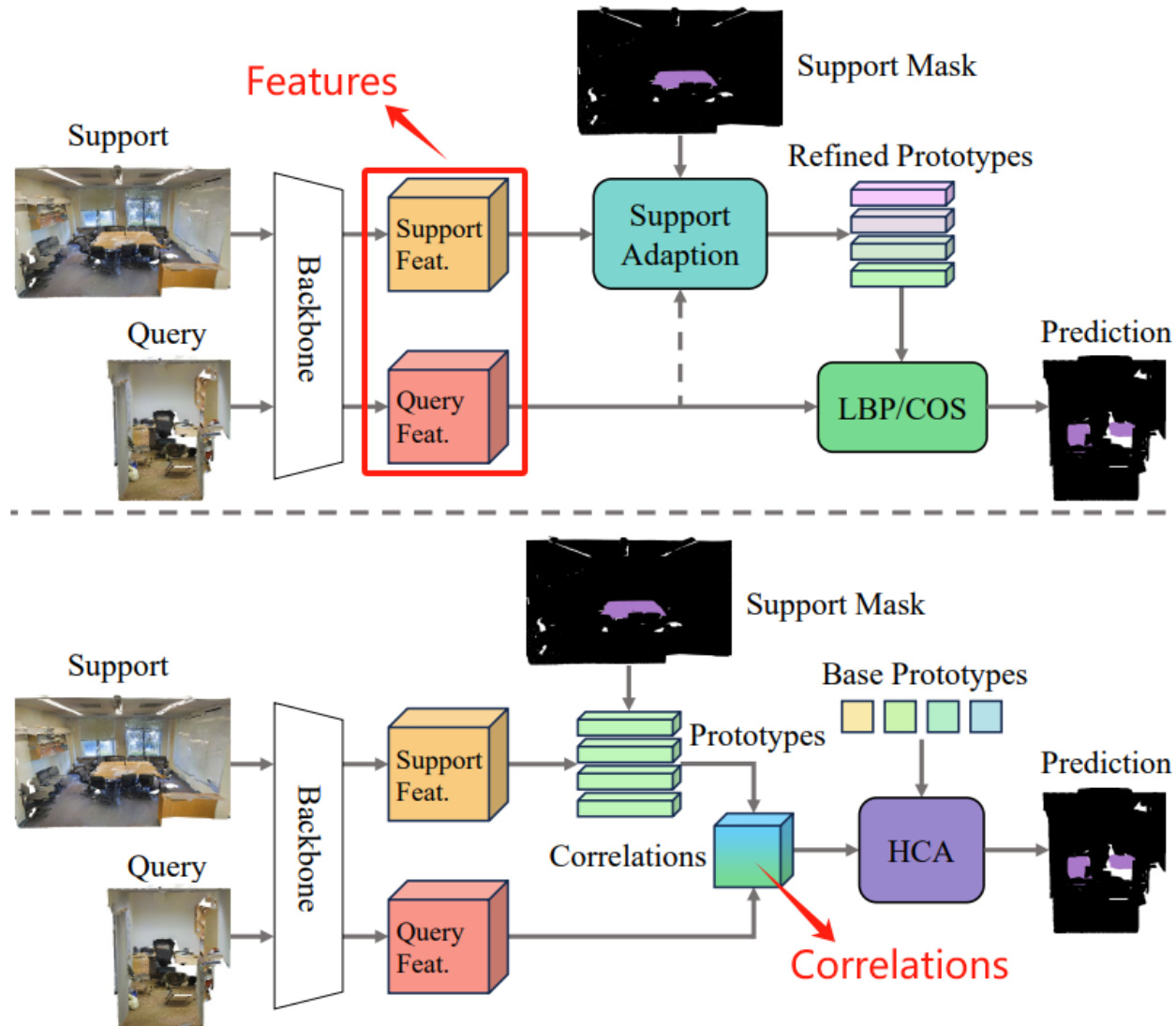
Issue 2: Sparse Point Distribution

- The current FSPCS input is constrained to only 2,048 points.
- These sparsely distributed, semantically limited inputs introduce significant ambiguities, hindering the model's capacity to exploit semantics in the scenes.



From left to right: (1) The original point cloud; (2) Ground truth of all categories; (3) Our corrected input with 20,480 points in a uniform distribution; (4) Input with 20,480 points in a biased distribution; (5) Input with 2,048 points in a uniform distribution; (6) Input with 2,048 points in a biased distribution, as adopted by previous works.

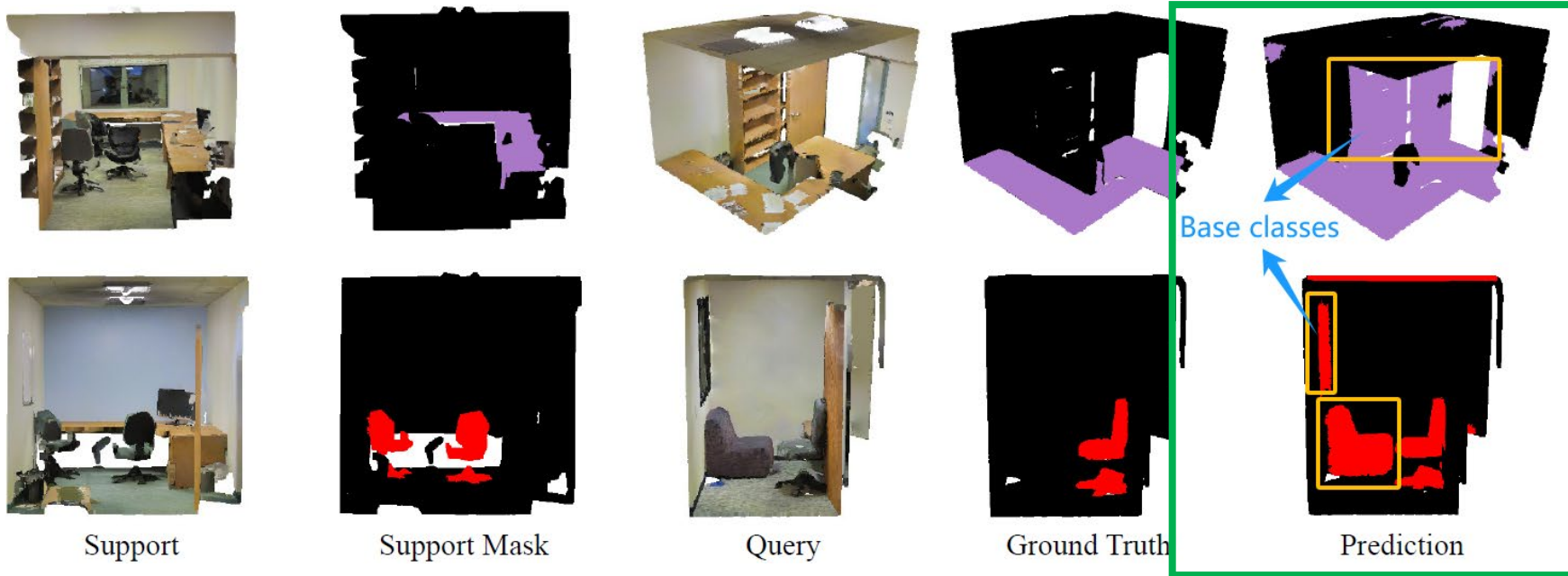
Motivation 1: Feature Optimization vs. Correlation Optimization



Top: Previous feature optimization; *Bottom:* Our correlation optimization

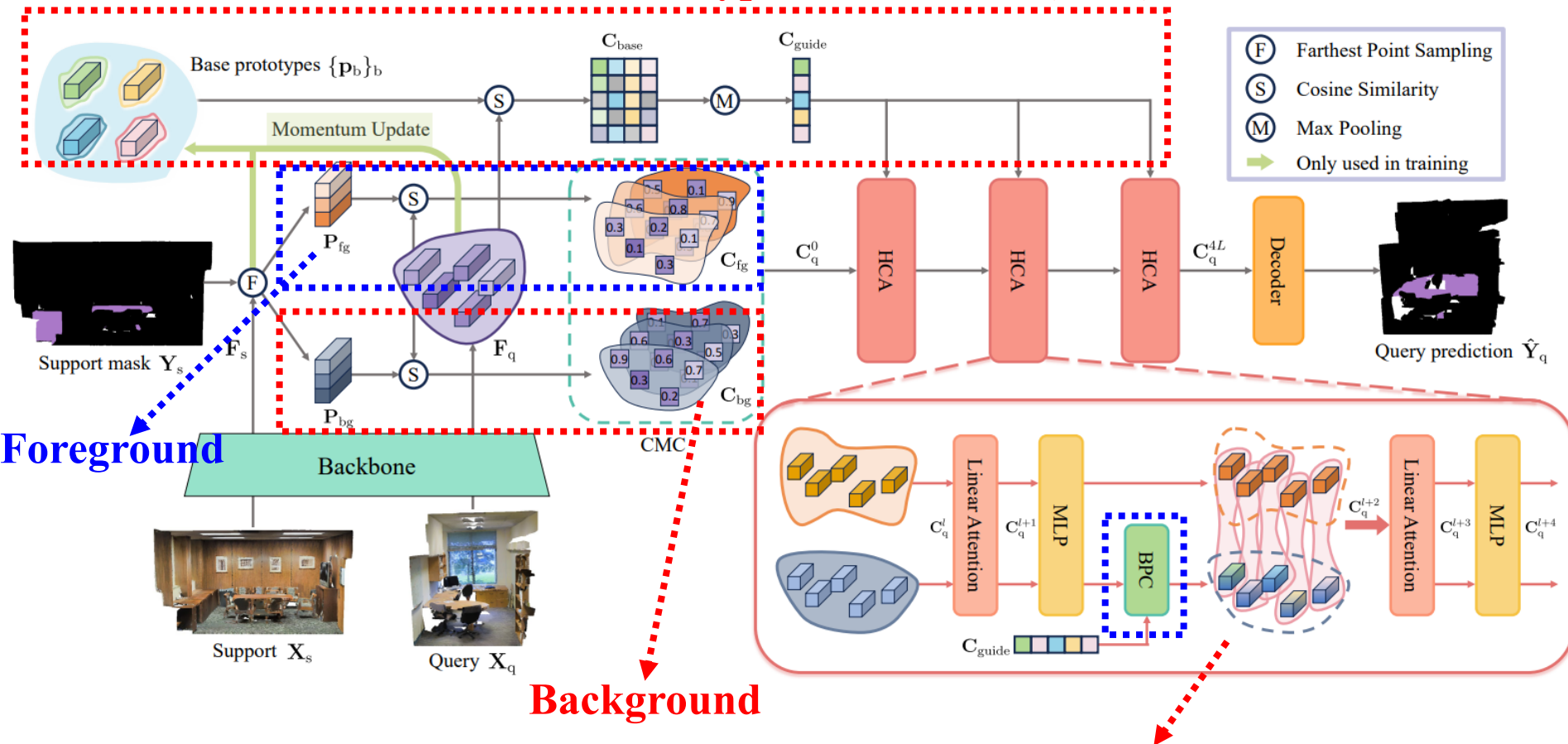
Motivation 2: Base Susceptibility Problem

- Within the meta-learning framework, models undergo training on *seen/base* classes and are evaluated on *unseen/novel* classes.
- These models tend to be susceptible to the base classes within test scenes, thereby hindering the accurate segmentation of novel classes.



Correlation Optimization Segmentation (COSeg)

Base Prototypes Calibration (BPC)



HCA: Hyper Correlation Augmentation, a carefully designed module for query-support correlation optimization. Please see the paper for details.

- Zhaochong An, Guolei Sun*, **Yun Liu***, Fayao Liu, Zongwei Wu, Dan Wang, Luc Van Gool, and Serge Belongie. "Rethinking Few-shot 3D Point Cloud Semantic Segmentation". **IEEE CVPR, 2024.**

Class-specific Multi-prototypical Correlation (CMC)

- Foreground prototypes $\mathbf{P}_{fg}$ and background prototypes $\mathbf{P}_{bg}$ are obtained through two steps: sample seeds in the coordinate space based on farthest point sampling, and then conduct point-to-seed clustering as follows:

$$\begin{aligned}\mathbf{P}_{fg} &= \mathcal{F}_{clus}(\mathbf{F}_s \odot \mathbf{Y}_s, \mathbf{S}_{fg}), \mathbf{S}_{fg} = \mathcal{F}_{fps}(\mathbf{L}_s \odot \mathbf{Y}_s), \\ \mathbf{P}_{bg} &= \mathcal{F}_{clus}(\mathbf{F}_s \odot \tilde{\mathbf{Y}}_s, \mathbf{S}_{bg}), \mathbf{S}_{bg} = \mathcal{F}_{fps}(\mathbf{L}_s \odot \tilde{\mathbf{Y}}_s)\end{aligned}$$

- We compute the cosine similarities of query points with respect to $\mathbf{P}_{fg}$ and $\mathbf{P}_{bg}$, and obtain the correlations:

$$\mathbf{C}_{fg} = \frac{\mathbf{F}_q \cdot \mathbf{P}_{fg}^\top}{\|\mathbf{F}_q\| \|\mathbf{P}_{fg}^\top\|}, \mathbf{C}_{bg} = \frac{\mathbf{F}_q \cdot \mathbf{P}_{bg}^\top}{\|\mathbf{F}_q\| \|\mathbf{P}_{bg}^\top\|}$$

- We concatenate $\mathbf{C}_{fg}$ and $\mathbf{C}_{bg}$ along the second dimension and project the last dimension back to D using an MLP, as follows:

$$\mathbf{C}_q^0 = \mathcal{F}_{mlp}(\mathbf{C}_{fg} \oplus \mathbf{C}_{bg}) \in \mathbb{R}^{N_Q \times N_C \times D}$$

Hyper Correlation Augmentation (HCA)

- We permute $\mathbf{C}_q^l$ with the class dimension as its first dimension and then compute linear attention across points:

$$\mathbf{C}_q^{l+1} = \mathcal{F}_{\text{lnatt}}(\mathcal{T}(\mathbf{C}_q^l)) \in \mathbb{R}^{N_C \times N_Q \times D}$$

- Following the attention layer, an MLP is applied:

$$\mathbf{C}_q^{l+2} = \mathcal{F}_{\text{mlp}}(\mathbf{C}_q^{l+1}) \in \mathbb{R}^{N_C \times N_Q \times D}$$

- We rearrange the dimensions and apply linear attention, given by:

$$\mathbf{C}_q^{l+3} = \mathcal{F}_{\text{lnatt}}(\mathcal{T}(\mathbf{C}_q^{l+2})) \in \mathbb{R}^{N_Q \times N_C \times D}$$

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Base Prototypes Calibration (BPC)

- During meta learning, we calculate the Masked Average Pooling (MAP) for each base class present in the current point clouds:

$$\mathbf{p}'_b = \mathcal{F}_{\text{pool}}(\mathbf{F}_{s/q} \odot \mathbf{Y}_{s/q}^b) \in \mathbb{R}^{1 \times D},$$
$$\mathbf{p}_b \leftarrow \mu \mathbf{p}_b + (1 - \mu) \mathbf{p}'_b$$

- We calculate the base correlations $\mathbf{C}_{\text{base}}$ between the query and base prototypes:

$$\mathbf{C}_{\text{base}} = \frac{\mathbf{F}_q \cdot \mathcal{I}(\{\mathbf{p}_b\}_{b=1}^{N_b})^\top}{\|\mathbf{F}_q\| \left\| \mathcal{I}(\{\mathbf{p}_b\}_{b=1}^{N_b})^\top \right\|} \in \mathbb{R}^{N_Q \times N_b}$$

- The background correlations are calibrated by $\mathbf{C}_{\text{guide}}$ before interacting with foreground correlations:

$$\mathbf{C}_{\text{guide}} = \mathcal{F}_{\text{max}}(\mathbf{C}_{\text{base}}) \in \mathbb{R}^{N_Q}$$
$$\mathbf{C}_q^{l+2}[1, \cdot, \cdot] = \mathcal{F}_{\text{fc}}(\mathbf{C}_q^{l+2}[1, \cdot, \cdot] \oplus \mathcal{D}(\mathbf{C}_{\text{guide}}))$$

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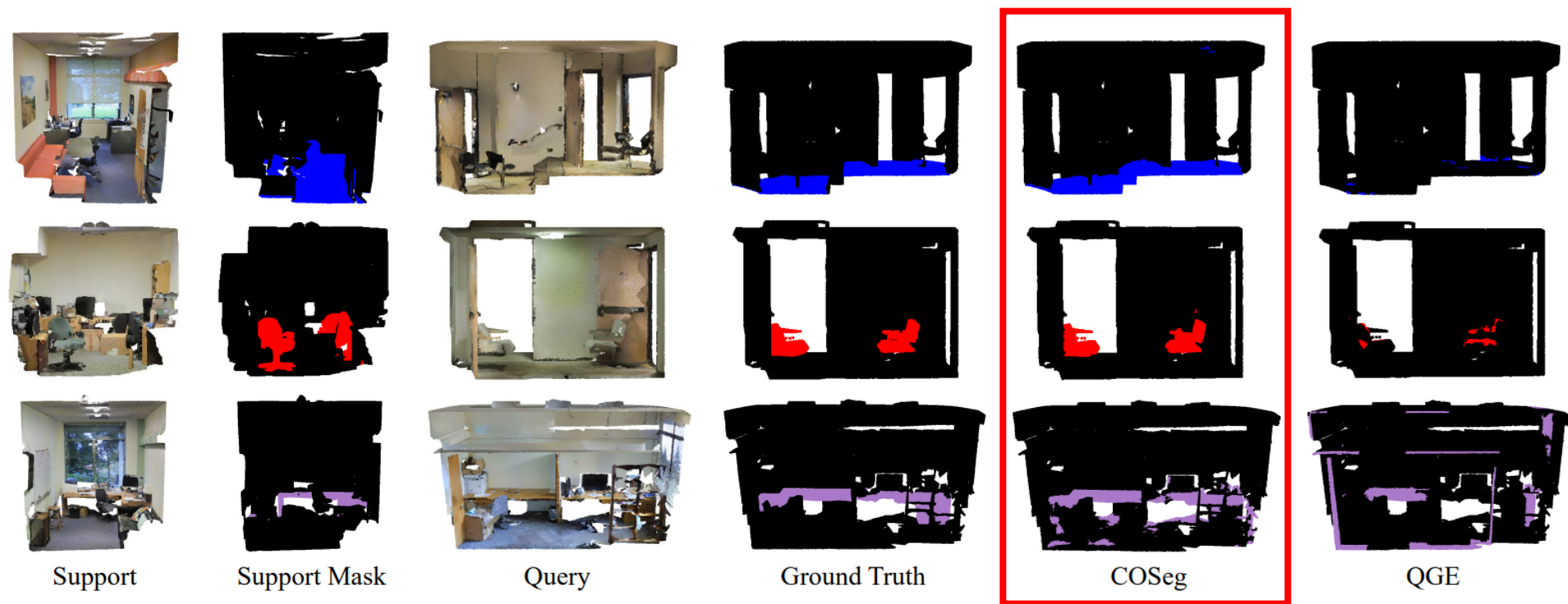
Quantitative Comparison

	Methods	1-way 1-shot			1-way 5-shot			2-way 1-shot			2-way 5-shot		
		S^0	S^1	mean	S^0	S^1	mean	S^0	S^1	mean	S^0	S^1	mean
S3DIS [1]	AttMPTI [56]	36.32	38.36	37.34	46.71	42.70	44.71	31.09	29.62	30.36	39.53	32.62	36.08
	QGE [29]	41.69	39.09	40.39	50.59	46.41	48.50	33.45	30.95	32.20	40.53	36.13	38.33
	QGPA [11]	35.50	35.83	35.67	38.07	39.70	38.89	25.52	26.26	25.89	30.22	32.41	31.32
	COSeg (ours)	46.31	48.10	47.21	51.40	48.68	50.04	37.44	36.45	36.95	42.27	38.45	40.36
ScanNet [7]	AttMPTI [56]	34.03	30.97	32.50	39.09	37.15	38.12	25.99	23.88	24.94	30.41	27.35	28.88
	QGE [29]	37.38	33.02	35.20	45.08	41.89	43.49	26.85	25.17	26.01	28.35	31.49	29.92
	QGPA [11]	34.57	33.37	33.97	41.22	38.65	39.94	21.86	21.47	21.67	30.67	27.69	29.18
	COSeg (ours)	41.73	41.82	41.78	48.31	44.11	46.21	28.72	28.83	28.78	35.97	33.39	34.68

Comparisons in the mIoU metric between our method and baselines in the new FS-PCS setting.

- Zhaochong An, Guolei Sun*, **Yun Liu***, Fayao Liu, Zongwei Wu, Dan Wang, Luc Van Gool, and Serge Belongie. “Rethinking Few-shot 3D Point Cloud Semantic Segmentation”. **IEEE CVPR, 2024.**

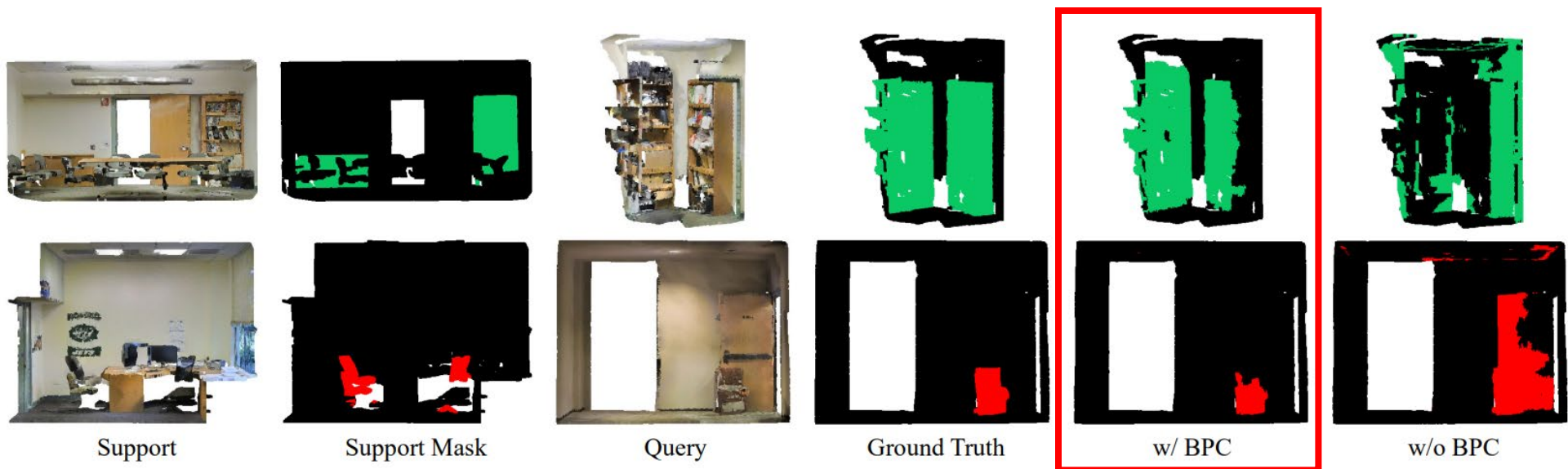
Qualitative Comparison



Qualitative comparisons between our proposed model COSeg and QGE (SOTA method). Each row, from top to bottom, represents the 1-way 1-shot task with the target category as floor (blue), chair (red), and table (purple), respectively.

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Qualitative Comparison



Visual comparisons between our models *with* BPC (*w/ BPC*) and *without* BPC (*w/o BPC*). Each row corresponds to the 1-way 1-shot task targeting bookcase (**green**) and chair (**red**), respectively, arranged from top to bottom.

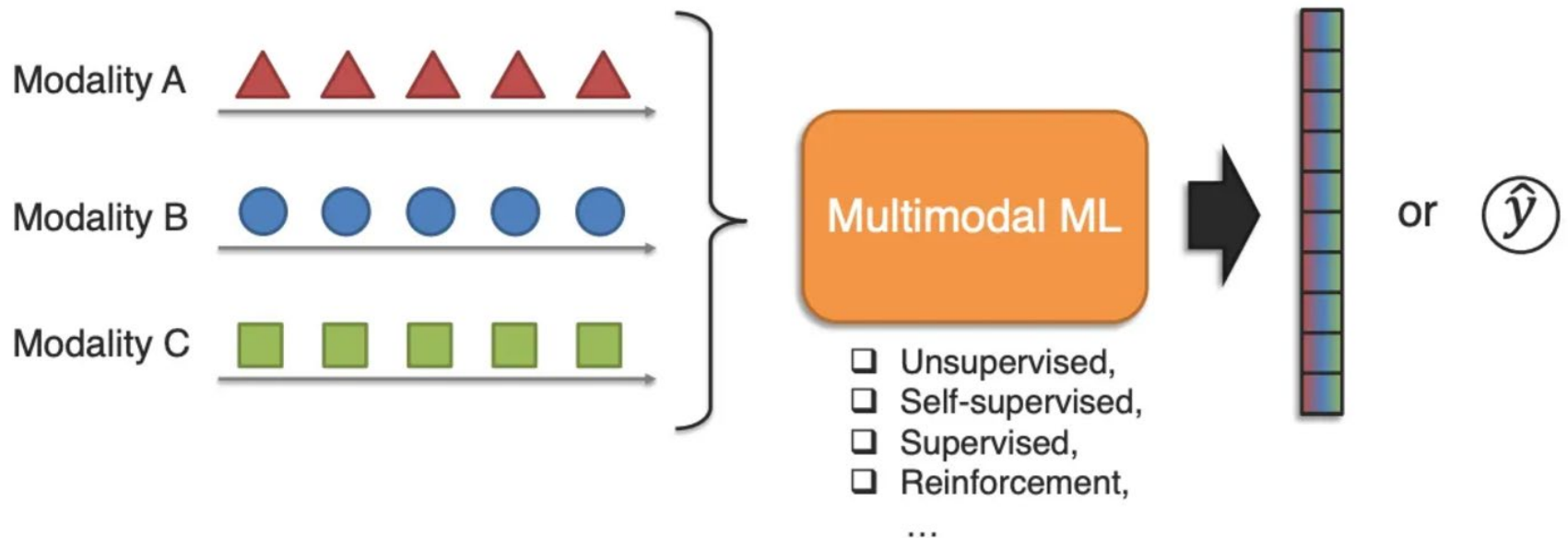
BPC: Base Prototypes Calibration

Code: <https://github.com/ZhaochongAn/COSeg>

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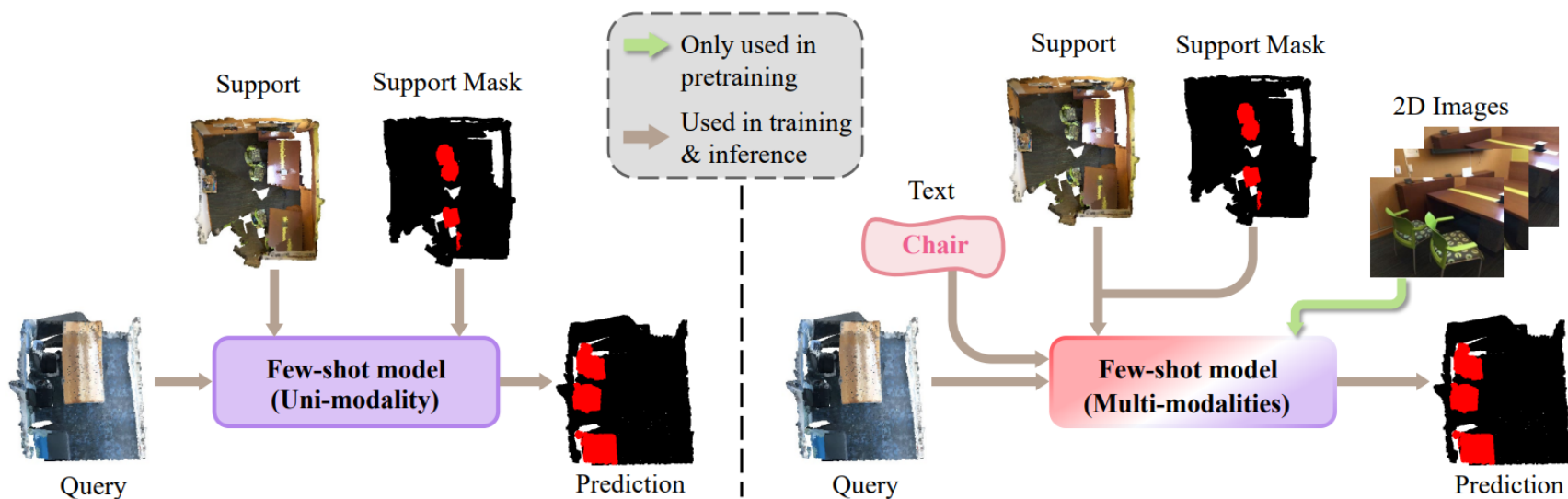
Multimodal Learning

- Multimodal learning is a type of deep learning that integrates and processes multiple types of data, referred to as modalities, such as text, audio, images, or video.
- Large multimodal models, such as Google Gemini and GPT-4o, have become increasingly popular since 2023, enabling increased versatility and a broader understanding of real-world phenomena.



Unimodal FS-PCS vs. Multimodal FS-PCS

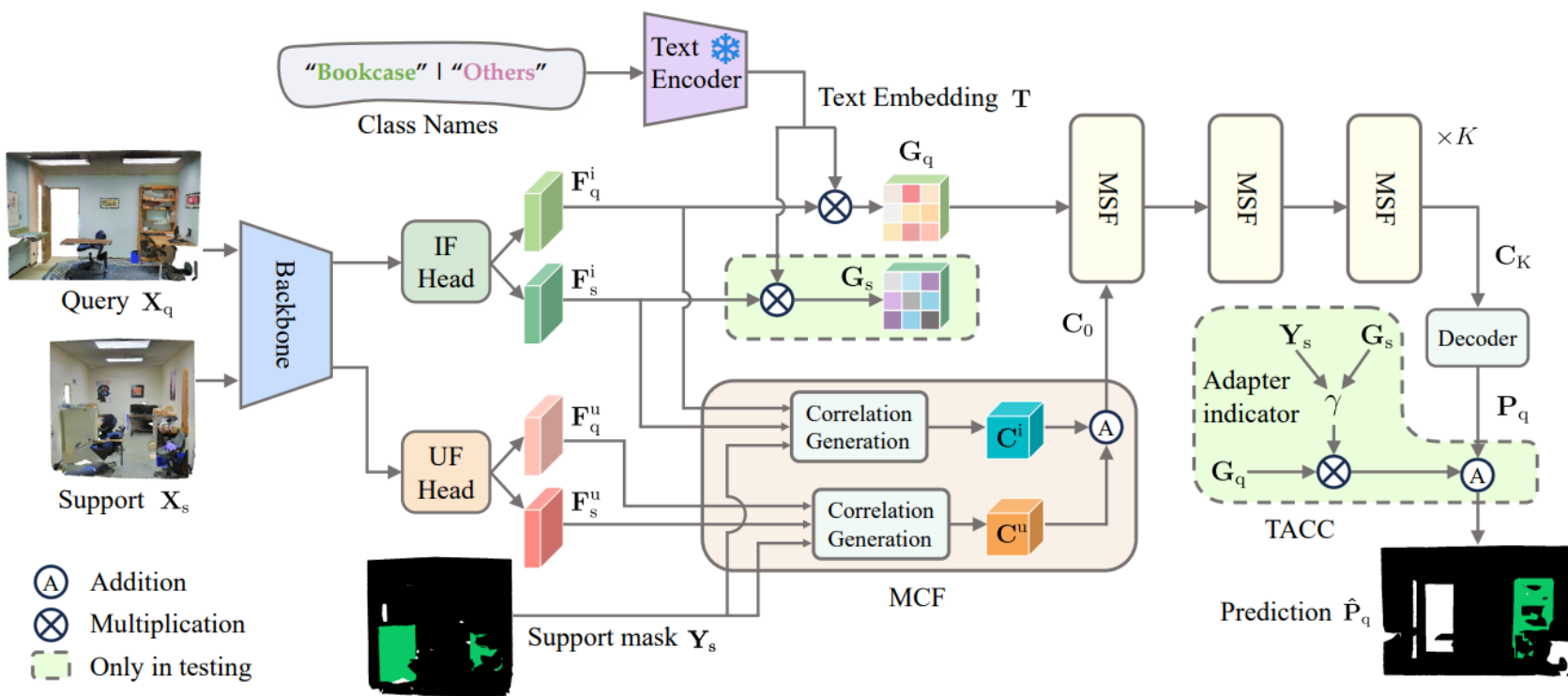
- Previous FS-PCS methods (left) only make use of point clouds as unimodal input. In contrast, our proposed model (right) utilizes **cost-free** multimodal information to improve FS-PCS by considering the textual modality of class names (*explicit*) and learning the simulated features of the 2D modality (*implicit*). During meta-learning and inference, the 2D modality is not needed.



- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Min Wu, Ming-Ming Cheng, Ender Konukoglu, and Serge Belongie. "Multimodality Helps Few-Shot 3D Point Cloud Semantic Segmentation". **ICLR 2025**.

Overall Architecture of Multimodal FS-PCS

- Given support and query point clouds, we first generate intermodal features $F_{s/q}^i$ from the IF head and unimodal features $F_{s/q}^u$ from the UF head. These features are then forwarded to the MCF module to generate initial multimodal correlations C_0 .



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Feature Extractors

- The IF head extracts intermodal features that are aligned with 2D visual features by exploiting the 2D modality, while the UF head focuses solely on the 3D point cloud modality.

$$\mathbf{F}_s^i = \mathcal{H}_{\text{IF}}(\mathbf{F}_s) \in \mathbb{R}^{N_S \times D_t}, \mathbf{F}_s^u = \mathcal{H}_{\text{UF}}(\mathbf{F}_s) \in \mathbb{R}^{N_S \times D},$$
$$\mathbf{F}_q^i = \mathcal{H}_{\text{IF}}(\mathbf{F}_q) \in \mathbb{R}^{N_Q \times D_t}, \mathbf{F}_q^u = \mathcal{H}_{\text{UF}}(\mathbf{F}_q) \in \mathbb{R}^{N_Q \times D}.$$

- In the pretraining, we employ a cosine similarity loss to minimize the distance between 3D point intermodal features and corresponding 2D pixel features. Then, we fix the backbone and IF head during meta-learning.
- We compute embeddings for the “background” and target classes using the LSeg text encoder, denoted as $\mathbf{T} = \{\mathbf{t}_0, \dots, \mathbf{t}_N\} \in \mathbb{R}^{N_C \times D_t}$

Multimodal Correlation Fusion (MCF)

- Prototypes are generated from the annotated support points for both intermodal and unimodal features. The correlations between the query points and these prototypes:

$$\mathbf{C}^i = \frac{\mathbf{F}_q^i \cdot \mathbf{P}_{\text{proto}}^{i\top}}{\|\mathbf{F}_q^i\| \|\mathbf{P}_{\text{proto}}^{i\top}\|}, \quad \mathbf{C}^u = \frac{\mathbf{F}_q^u \cdot \mathbf{P}_{\text{proto}}^{u\top}}{\|\mathbf{F}_q^u\| \|\mathbf{P}_{\text{proto}}^{u\top}\|}$$

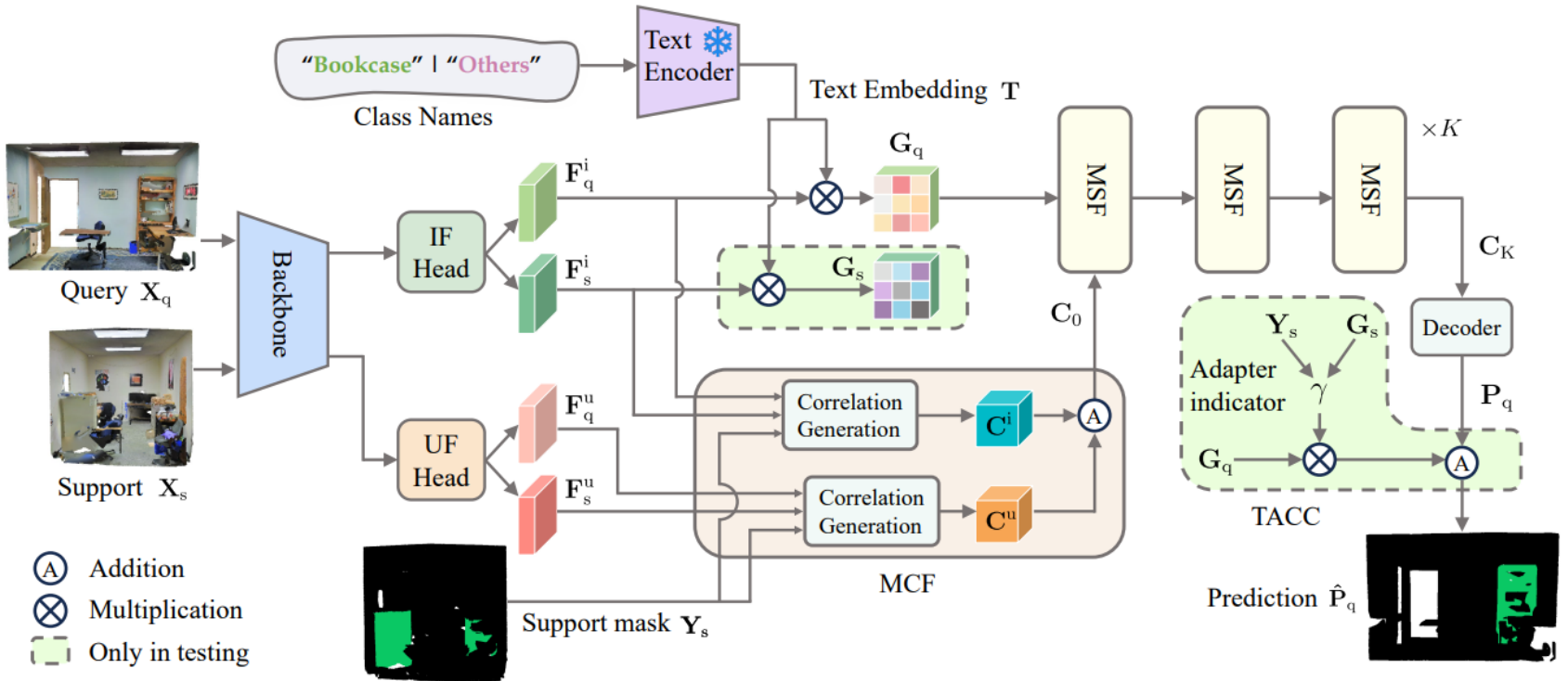
- The MCF module transforms these correlations using two linear layers and then combines them to obtain the aggregated multimodal correlation $\mathbf{C}_0$, as follows:

$$\mathbf{C}_0 = \mathcal{F}_{\text{lin}}(\mathbf{C}^i) + \mathcal{F}_{\text{lin}}(\mathbf{C}^u), \quad \mathbf{C}_0 \in \mathbb{R}^{N_Q \times N_C \times D}$$

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Overall Architecture of Multimodal FS-PCS

- For exploiting the alignment between intermodal features F_q^i and text embeddings T , we use their affinity G_q as the informative textual semantic guidance to refine the multimodal correlations in the MSF modules. Finally, we propose the TACC, a parameter-free module that adaptively calibrates predictions during test time to effectively mitigate the base bias issue.



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Multimodal Semantic Fusion (MSF)

- We first compute the similarity between the query intermodal features and text embeddings to generate semantic guidance:

$$\mathbf{G}_q = \mathbf{F}_q^i \cdot \mathbf{T}^\top$$

- Point-category weights to consider varying importance between visual and textual modalities are dynamically computed as follows:

$$\mathbf{W}_q = \mathcal{F}_{\text{mlp}}(\mathcal{F}_{\text{expand}}(\mathbf{G}_q) \oplus \mathbf{C}_k), \quad \mathbf{W}_q \in \mathbb{R}^{N_Q \times N_C \times 1}$$

- The semantic guidance $\mathbf{G}_q$, weighted by $\mathbf{W}_q$, is aggregated into the correlation input $\mathbf{C}_k$:

$$\begin{aligned} \mathbf{C}'_k &= \mathbf{G}_q \odot \mathbf{W}_q + \mathbf{C}_k, \\ \mathbf{C}_{k+1} &= \mathcal{F}_{\text{mlp}}(\mathcal{F}_{\text{attention}}(\mathbf{C}'_k)) \end{aligned}$$

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Test-time Adaptive Cross-modal Calibration (TACC)

- We propose an adaptive combination of the semantic guidance $\mathbf{G}_q$ and the prediction $\mathbf{P}_q$ through an adaptive indicator γ :

$$\hat{\mathbf{P}}_q = \gamma \mathbf{G}_q + \mathbf{P}_q$$

- Using the support intermodal features $\mathbf{F}$ and the text embeddings $\mathbf{T}$, we compute $\mathbf{G}_s$, which is then used to generate the predicted labels $\mathbf{P}_s$. With the available support labels $\mathbf{Y}_s$ in each episode, the quality of $\mathbf{G}_s/\mathbf{G}_q$ is quantified by comparing the predicted labels $\mathbf{P}_s$ to $\mathbf{Y}_s$ using the Intersection-over-Union (IoU) score.

$$\gamma = \frac{\sum_i \mathbf{1}_{\{\mathbf{P}_s(i)=1 \wedge \mathbf{Y}_s(i)=1\}}}{\sum_i \mathbf{1}_{\{\mathbf{P}_s(i)=1 \vee \mathbf{Y}_s(i)=1\}}}, \quad \mathbf{P}_s[i] = \arg \max(\mathbf{G}_s[i, :]), \quad \mathbf{G}_s = \mathbf{F}_s^i \cdot \mathbf{T}^\top$$

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Quantitative Comparison

Methods	1-way 1-shot			1-way 5-shot			2-way 1-shot			2-way 5-shot		
	S^0	S^1	Mean	S^0	S^1	Mean	S^0	S^1	Mean	S^0	S^1	Mean
AttMPTI (Zhao et al., 2021)	36.32	38.36	37.34	46.71	42.70	44.71	31.09	29.62	30.36	39.53	32.62	36.08
QGE (Ning et al., 2023)	41.69	39.09	40.39	50.59	46.41	48.50	33.45	30.95	32.20	40.53	36.13	38.33
QGPA (He et al., 2023)	35.50	35.83	35.67	38.07	39.70	38.89	25.52	26.26	25.89	30.22	32.41	31.32
COSeg (An et al., 2024)	46.31	48.10	47.21	51.40	48.68	50.04	37.44	36.45	36.95	42.27	38.45	40.36
COSeg [†] (An et al., 2024)	47.17	48.37	47.77	50.93	49.88	50.41	37.15	38.99	38.07	42.73	40.25	41.49
MM-FSS (ours)	49.84	54.33	52.09(+4.3)	51.95	56.46	54.21(+3.8)	41.98	46.61	44.30(+6.2)	46.02	54.29	50.16(+8.7)

Table 1: **Quantitative comparison with previous methods in mIoU (%) on the S3DIS dataset.** There are four few-shot settings: 1/2-way 1/5-shot. S^0/S^1 refers to using the split i for evaluation, and ‘Mean’ represents the average mIoU on both splits. The best results are highlighted in **bold**.

Methods	1-way 1-shot			1-way 5-shot			2-way 1-shot			2-way 5-shot		
	S^0	S^1	mean	S^0	S^1	Mean	S^0	S^1	Mean	S^0	S^1	Mean
AttMPTI (Zhao et al., 2021)	34.03	30.97	32.50	39.09	37.15	38.12	25.99	23.88	24.94	30.41	27.35	28.88
QGE (Ning et al., 2023)	37.38	33.02	35.20	45.08	41.89	43.49	26.85	25.17	26.01	28.35	31.49	29.92
QGPA (He et al., 2023)	34.57	33.37	33.97	41.22	38.65	39.94	21.86	21.47	21.67	30.67	27.69	29.18
COSeg (An et al., 2024)	41.73	41.82	41.78	48.31	44.11	46.21	28.72	28.83	28.78	35.97	33.39	34.68
COSeg [†] (An et al., 2024)	41.95	42.07	42.01	48.54	44.68	46.61	29.54	28.51	29.03	36.87	34.15	35.51
MM-FSS (ours)	46.08	43.37	44.73(+2.7)	54.66	45.48	50.07(+3.5)	43.99	34.43	39.21(+10.2)	48.86	39.32	44.09(+8.6)

Table 2: **Quantitative comparison with previous methods in mIoU (%) on the ScanNet dataset.**

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Qualitative Comparison

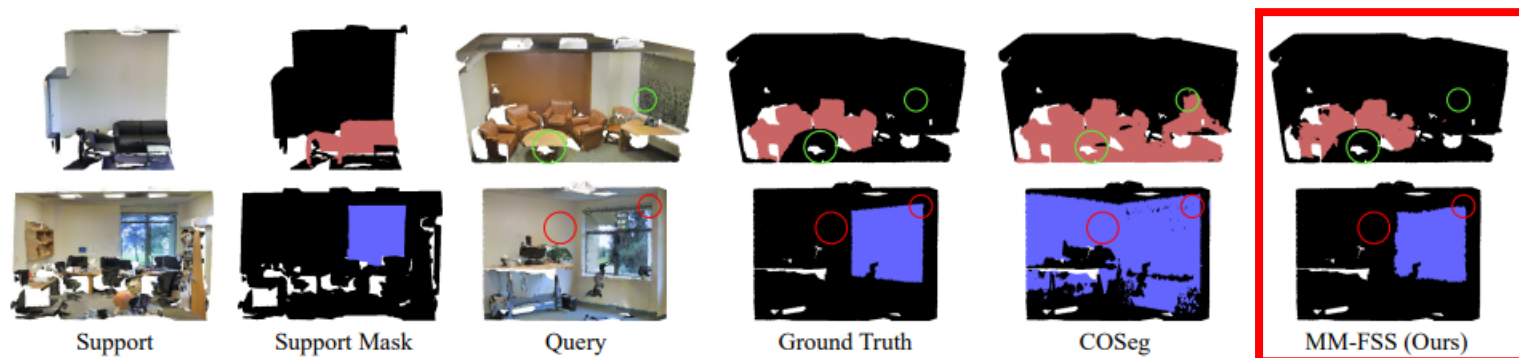


Figure 3: **Qualitative comparison between COSeg and our proposed MM-FSS in the 1-way 1-shot setting on the S3DIS dataset.** The target classes in the first and second rows are **sofa** and **window**, respectively. Important areas are marked with circles.

Code: <https://github.com/ZhaochongAn/Multimodality-3D-Few-Shot>

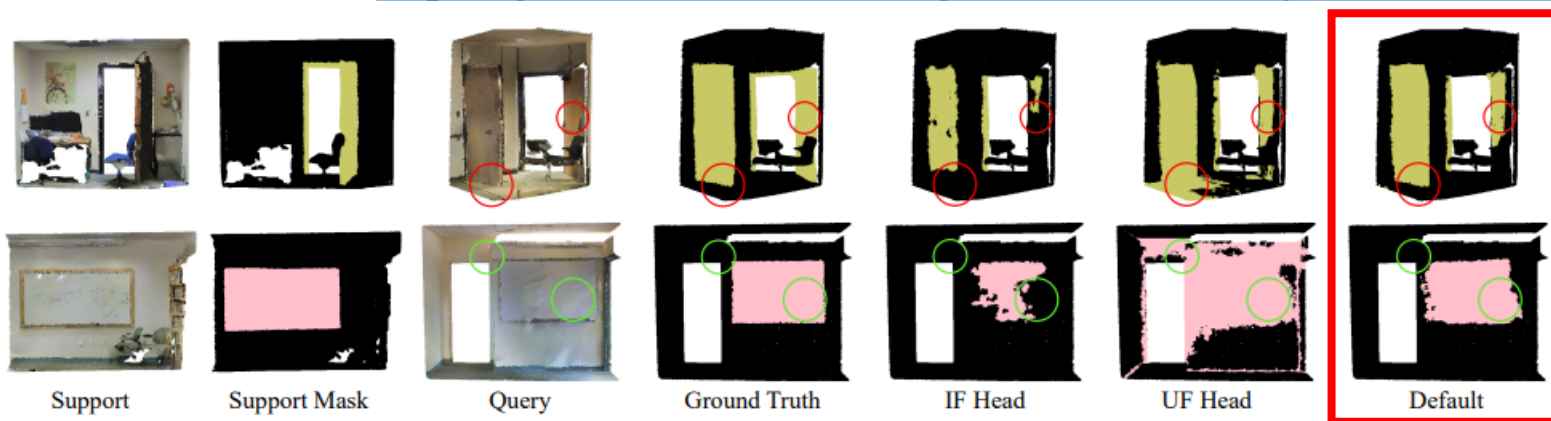
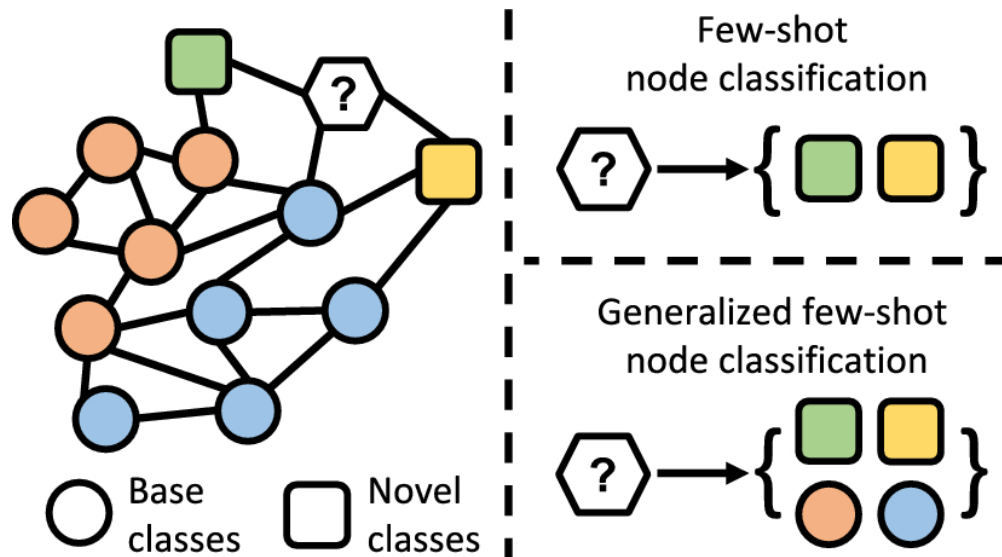


Figure 4: **Qualitative comparison of predictions from each head and our final prediction using TACC (Default) in the 1-way 1-shot setting on the S3DIS dataset.** The target classes in the first and second rows are **door** and **board**, respectively.

- Zhaochong An, Guolei Sun*, Yun Liu*, Runjia Li, Min Wu, Ming-Ming Cheng, Ender Konukoglu, and Serge Belongie. “Multimodality Helps Few-Shot 3D Point Cloud Semantic Segmentation”. **ICLR 2025**.

Few-shot Seg. vs. Generalized Few-shot Seg.

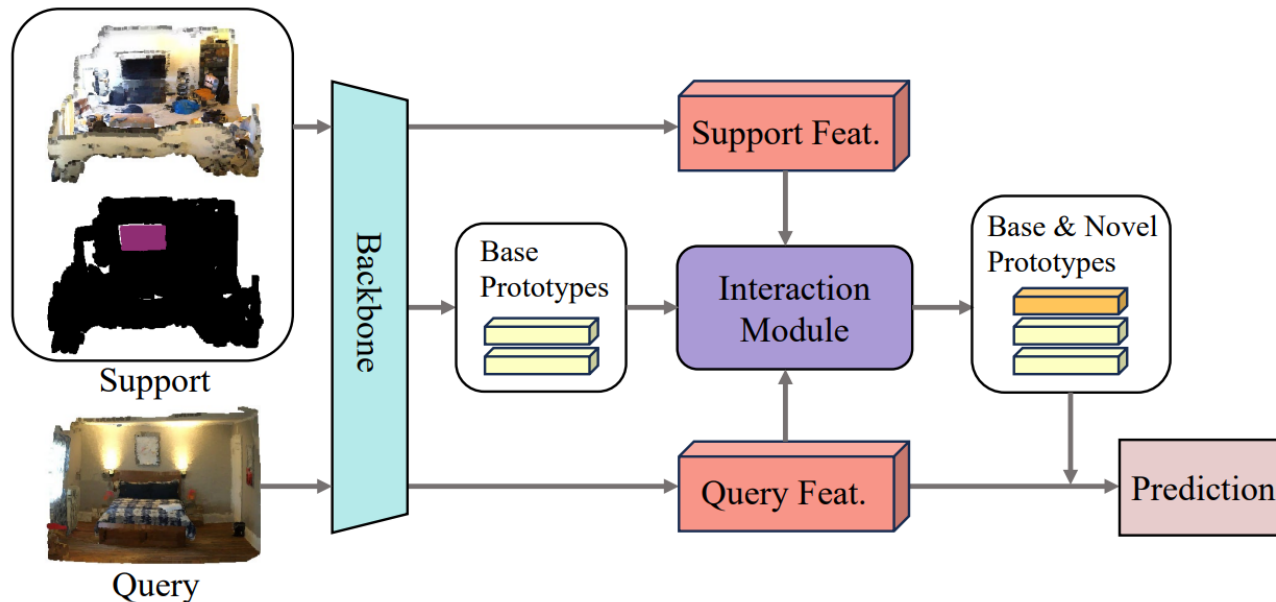
- Few-shot segmentation requires additional support samples for each novel class at inference and **only predicts novel classes**, ignoring base classes.
- Generalized Few-shot segmentation directly segments **both base and novel classes** after few-shot adaptation, making it more practical for real-world applications.
- **GFS-PCS: Generalized Few-shot 3D Point Cloud Segmentation**



- Zhe Xu, Kaize Ding, Yu-Xiong Wang, Huan Liu, and Hanghang Tong. “Generalized few-shot node classification: toward an uncertainty-based solution”. **Knowledge and Information Systems** 2024.

Challenge in GFS-PCS

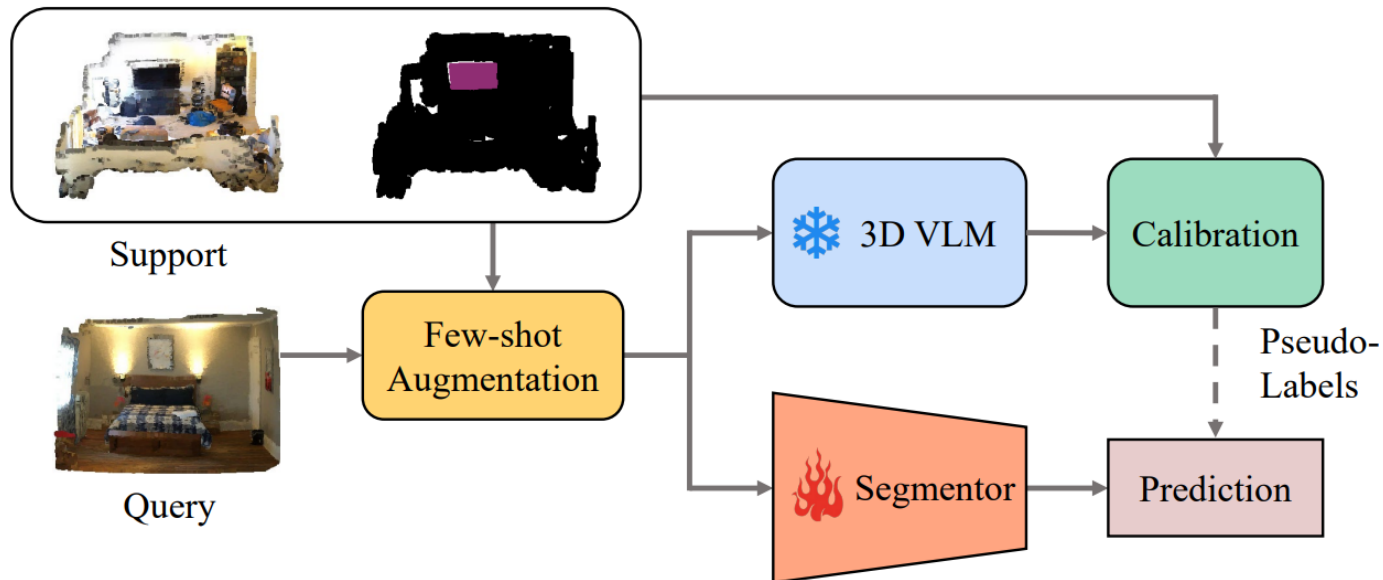
- Prior work primarily enhances prototypes through interaction modules that integrate support/query features, making predictions based on refined prototypes. However, they are **limited by the sparse knowledge from few-shot samples**.



- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

GFS-PCS with 3D Vision-Language Models (3D VLMs)

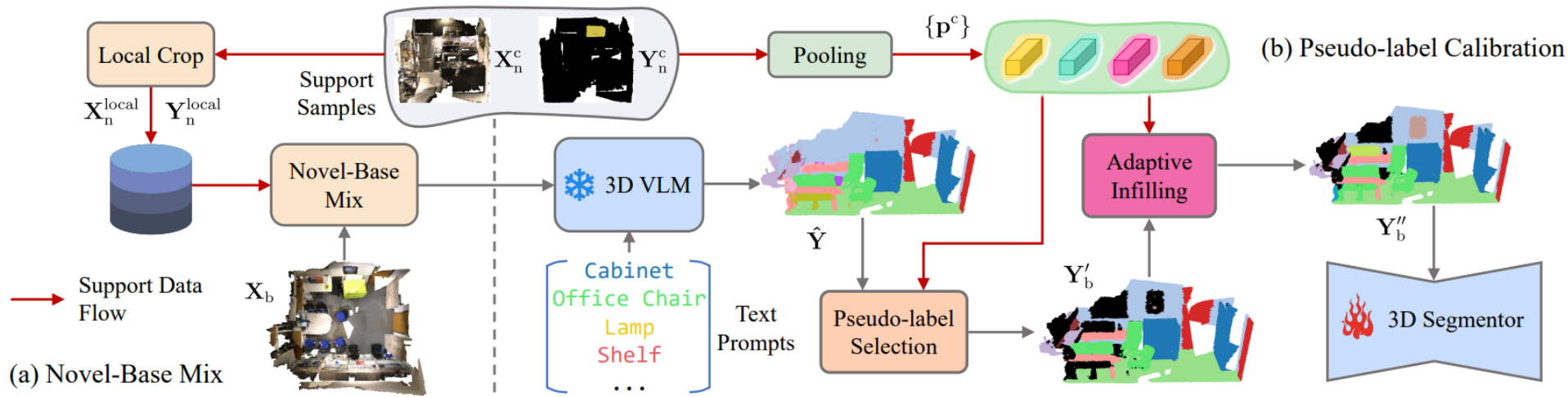
- Our framework addresses this limitation by leveraging the extensive open-world knowledge from 3D VLMs through pseudo-labels. We mitigate the noise inherent in 3D VLMs by calibrating their raw pseudo-labels with precise few-shot samples, thereby effectively expanding novel class knowledge while ensuring reliability.



- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

GFS-PCS with 3D Vision-Language Models (3D VLMs)

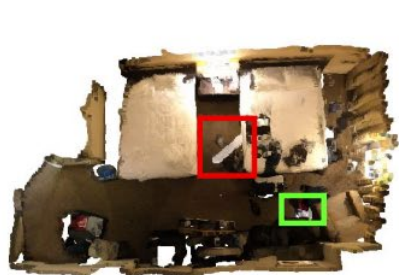
- Given an input point cloud $\mathbf{X}_b$, we apply a novel-base mix to embed support samples into the training scene while preserving essential context. The scene is then processed by a 3D VLM, using all class names as prompts to generate raw predictions $\hat{\mathbf{Y}}$. Leveraging support prototypes $\{\mathbf{p}^c\}$, the raw predictions undergo pseudo-label selection to filter out noisy regions, followed by adaptive infilling to label the filtered, unlabeled areas, yielding refined supervision $\mathbf{Y}_b''$ for training the 3D segmentor.



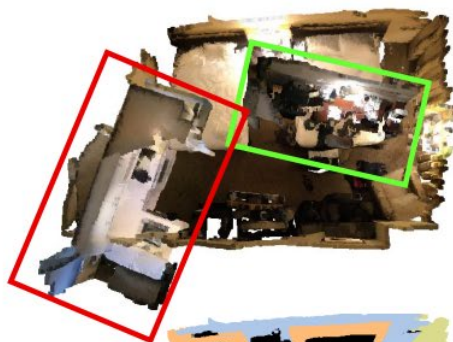
- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

GFS-PCS with 3D Vision-Language Models (3D VLMs)

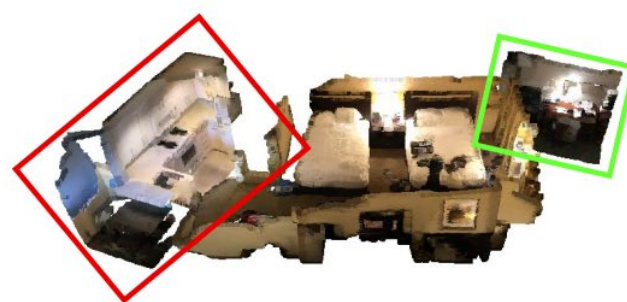
- Visual illustration of mixing strategies. The red and green boxes represent the two novel samples mixed into the scene.



Instance Mix



Mix3D

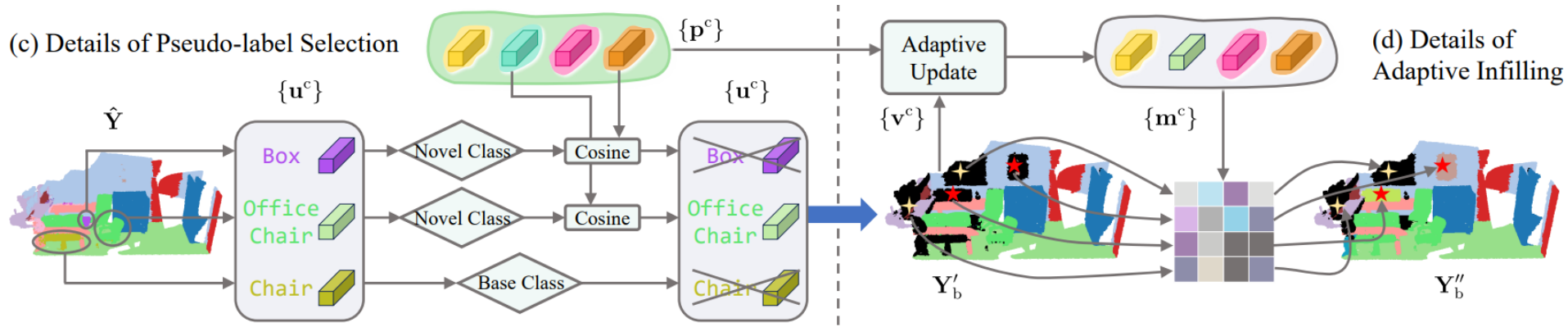


Novel-Base Mix (Ours)

- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

GFS-PCS with 3D Vision-Language Models (3D VLMs)

- (c) (d) illustrate the details of the pseudo-label selection and adaptive infilling processes, respectively.



- We further propose two new and more challenging evaluation benchmarks based on ScanNet200 and ScanNet++ datasets.

Dataset	Base	Novel	Max (F)	Min (F)	Max (P)	Min (P)
S3DIS	7	6	185	29	59,929	30,013
ScanNet	13	6	411	133	4,479	1,148
ScanNet200	12	45	733	102	12,641	279
ScanNet++	12	18	143	82	84,375	604

- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

Quantitative Comparison

Method	5-shot				1-shot			
	mIoU-B	mIoU-N	mIoU-A	HM	mIoU-B	mIoU-N	mIoU-A	HM
Fully Supervised	68.70	39.32	45.51	50.02	68.70	39.32	45.51	50.02
PIFS [4]	28.78	3.82	9.07	6.71	17.84	2.87	6.02	4.88
attMPTI [77]	37.13	4.99	11.76	8.79	54.84	3.28	14.14	6.17
COSeg [2]	57.67	5.21	16.25	9.54	47.03	4.03	13.09	7.42
GW [66]	59.28	8.30	19.03	14.55	55.23	6.47	16.74	11.56
GFS-VL (ours)	67.57	31.67	39.23	43.12	68.48	29.18	37.45	40.92

Table 2. Comparisons of our method with baselines on the new ScanNet200 benchmark. The best results are highlighted in **bold**.

Method	5-shot				1-shot			
	mIoU-B	mIoU-N	mIoU-A	HM	mIoU-B	mIoU-N	mIoU-A	HM
Fully Supervised	65.45	37.24	48.53	47.47	65.45	37.24	48.53	47.47
PIFS [4]	39.98	5.74	19.44	10.03	36.66	4.95	17.63	8.71
attMPTI [77]	55.89	4.19	24.87	7.78	53.16	3.55	23.40	6.66
COSeg [2]	59.34	6.96	27.91	12.45	58.49	6.24	27.14	11.26
GW [66]	51.35	11.03	27.16	18.15	46.71	6.63	22.66	11.59
GFS-VL (ours)	60.05	21.66	37.02	31.82	61.39	19.42	36.21	29.47

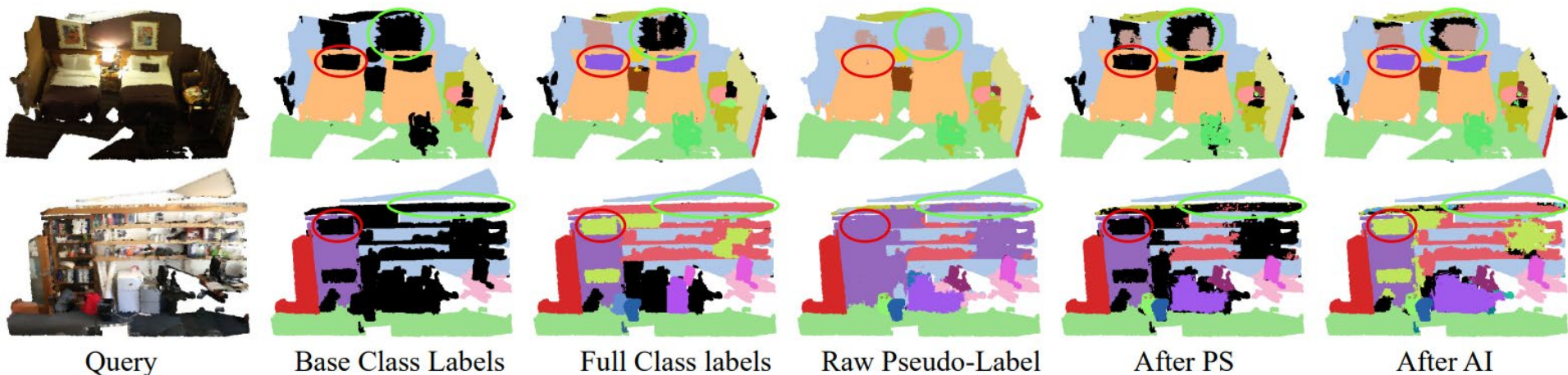
Table 3. Comparisons of our method with baselines on the new ScanNet++ benchmark. The best results are highlighted in **bold**.

Method	5-shot				1-shot			
	mIoU-B	mIoU-N	mIoU-A	HM	mIoU-B	mIoU-N	mIoU-A	HM
Fully Supervised	78.71	60.37	72.91	68.33	78.71	60.37	72.91	68.33
attMPTI [77]	16.31	3.12	12.35	5.21	12.97	1.62	9.57	2.88
PIFS [4]	35.14	3.21	25.56	5.88	35.80	2.54	25.82	4.75
CAPL [56]	38.22	14.39	31.07	20.88	38.70	10.59	30.27	16.53
GW [66]	40.18	18.58	33.70	25.39	40.06	14.78	32.47	21.55
GFS-VL (ours)	78.30	51.22	69.75	61.91	78.56	49.72	69.45	60.88

Table 4. Comparisons of our method with baselines on the old ScanNet benchmark. The best results are highlighted in **bold**.

Qualitative Comparison

- Visualization of the improvements in pseudo-label quality after applying Pseudo-label Selection (PS) and Adaptive Infilling (AI). Note that AI effectively discovers missed novel classes in the red circles and completes partial pseudo-labels in the green circles.



Code: <https://github.com/ZhaochongAn/GFS-VL>

- Zhaochong An, Guolei Sun*, **Yun Liu***, Runjia Li, Junlin Han, Ender Konukoglu, and Serge Belongie. “Generalized Few-shot 3D Point Cloud Segmentation with Vision-Language Model”. **IEEE CVPR, 2025.**

https://yun-liu.github.io/materials/Slides_FS-PCS.pdf

Thank you!